

Banana and its by-products: A comprehensive review on its nutritional composition and pharmacological benefits

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Funding information

None

Abstract

Bananas are widely popular as a key member of the Musaceae family and also considered a rich source of several nutrients, especially bioactive compounds. Besides, bananas are extensively grown in tropical and subtropical regions and are easily available for various use cases, that is, food industry and health benefits. Other than banana, its by-products such as peel, pseudo-stems, leaves, and blossoms are also rich in several nutrients, for example, carbohydrates, protein, dietary fiber, vitamins, and so on. Moreover, their consumption intends to provide several therapeutic benefits, particularly the dietary fiber and phenolic compounds. Furthermore, bananas and their by-products have been found to possess antimicrobial, anticancer, and antioxidant activities. In spite of countless benefits, these residues are often discarded as waste. Observing these benefits, the current review focuses on the broad range of bioactive chemical and pharmacological elements in bananas and their by-products. Also, this work focuses on their use in several food industries. As a result of the findings, the presented review reveals several innovative aspects of bananas and their products which can be utilized as a sustainable source of income for the agriculture industry.

KEYWORDS

antimicrobial, antioxidant, banana and its by-products, commercial utilization, nutrients, therapeutic benefits

1 | INTRODUCTION

In botanical terminology, “banana” refers to an ever-green monocot, perennial, large, subtropical herb belonging to the family Musaceae and the genus *Musa*. Globally, more than a thousand banana types are cultivated and produced. There are three species that are commonly cultivated among these species, namely *Musa cavendishi*, *Musa paradisiaca*, and *Musa sapientum*. Recent statistics published by the Food and Agriculture Organization (FAO) indicate that India produces 30,460,000 tons of bananas annually, followed by China and Indonesia, as illustrated in Figure 1 (Mostafa, 2021).

Among the continents that produced the most bananas, Asia was the leading producer, followed by

Africa, South America, Central America, and so on. Asia accounted for 68041.17 tonnes of bananas, Africa accounted for 22941.96 tonnes, South America accounted for 19421.82 tonnes, Central America accounted for 10188.53 tonnes, the Caribbean accounted for 2017 tonnes, Oceania accounted for 1696.17 tonnes, Europe accounted for 668.17 tonnes, and North America accounted for 2.78 tonnes. (FAOSTAT, 2021). As the world's largest producer of mango (45%), papaya (37%), bananas (29%), pomegranates, sapota, and acid lime, India also records the highest grape productivity (21.6 t/ha). More than 60% of the total banana production in the world is produced by India, China, the Philippines, Brazil, and Ecuador (Nayak & Nahar, 2018). Bananas, being the most highly produced and consumed fruit in

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India, account for approximately 33.4% of total fruit production in the country. A major producer of bananas in the world, India occupies approximately 15% of the world's banana areas and produces about 29% of the world's total banana production. Compared to 21.20 Mt./ha for the world, India's banana productivity is 37.90 Mt/ha (National Horticultural Board [NHB], 2015).

In India, the states showing the highest banana production include Gujrat, Maharashtra, and Tamil Nadu with total productivity of 66, 58.2, and 47.9 tons/ha. Other states with banana production include Andhra Pradesh, Uttar Pradesh, West Bengal, Karnataka, Madhya Pradesh, Bihar, Odisha, and Assam. Across the country, bananas are produced at an average rate of 38.25 tons per hectare (Uma & Kumar, 2020). They are extensively grown in tropical and subtropical regions and require a uniform warm and rainy climate year-round. They are considered a semi-perennial crop with a crop cycle nearly a year long under optimal conditions and even longer with lower temperatures or a more erratic water supply (Calberto et al., 2015).

Banana trees take 8–13 months to grow and yield fruit. The ideal time for harvesting bananas and the harvest interval may differ according to the variety and climatic zone. The banana vine can be harvested when the fruit is fully grown and turns green in color. Additionally, mature green banana fruits are artificially ripened in controlled environmental conditions or in a natural environment with ambient temperatures. The ripening process involves the transformation of banana bunches from brown or mature-yellow color into dark yellow skin with several features of the fruit namely the consistency of pulp, flavors, and transition of starch to sugar (Qamar & Shaikh, 2018).

Highlights

- Description of banana and its by-products like peel, pseudo-stems, leaves, and blossoms.
- Nutritional value, bioactive compounds, and therapeutic properties of banana and by-products.
- Utilization of these by-products in the food industry as functional ingredients.

Bananas are considered to be a valuable dietary component due to their high content of dietary fiber, essential vitamins, and minerals. Banana peel makes up around half of the total mass of the fruit (35%–50%) out of all by-products of the banana such as pseudo-stems, leaves, and blossoms (Ahmadi et al., 2019; Gomes et al., 2022). The banana pseudo-stem (BPS) grows around 6–7.6 m tall, bears flowers, leaves, and fruits. It makes up the majority of plant biomass, which is typically burned and discarded (Al-Mqbali & Hossain, 2019). The increasing demand for the utilization of BPS in recent years can be attributed to its high availability of numerous nutrients such as protein, carbs, and dietary fiber, which contribute to its enhanced nutritional benefits (Ho et al., 2017). Banana leaves are large, flexible, waterproof, and around 2.7×0.6 m in size. Due to their substantial waxy surface, they are appropriate for food packaging and food serving (Ahmadi et al., 2019).

Banana peels (BP), which make up approximately 40% of the banana fruit, are typically wasted and contribute to a number of environmental issues, such as they may lead to

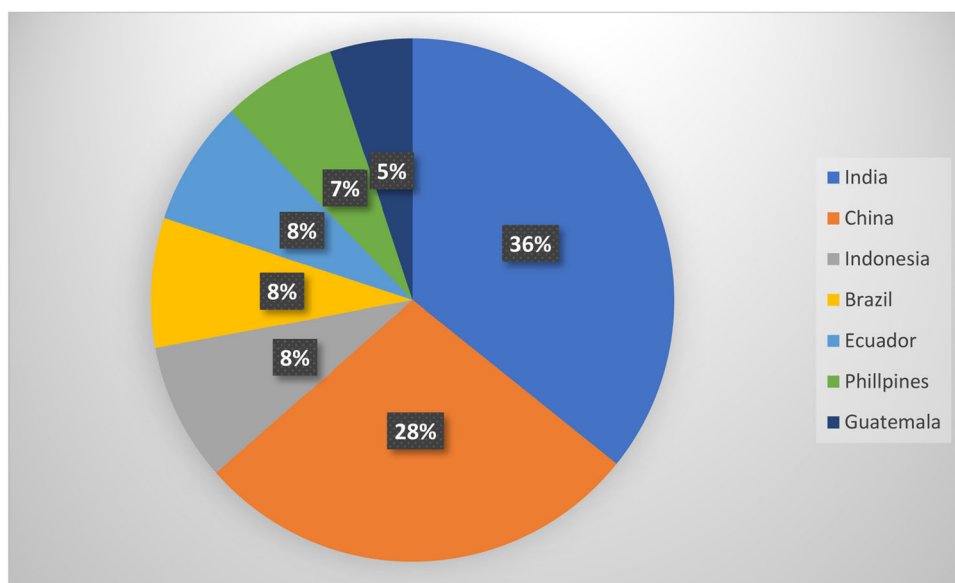


FIGURE 1 The top banana-producing countries and their yield (thousand ton/year). Source: FAOSTAT, 2019; Mostafa, 2021.

the production of harmful gases namely ammonia and hydrogen sulfide in the atmosphere (Kamel et al., 2017). Additionally, BP possesses a high concentration of phytochemicals, ranging from 0.90 to 3.0 g/100 g dry weight, including β -carotene, vitamin C, tocopherol, and gallic acid (Khawas & Deka, 2017). The banana blossom, commonly employed as a dietary vegetable, represents one of the lesser-utilized components of the banana plant (Suprayitno et al., 2022). It contains proteins, dietary fiber, vitamins, flavonoids (especially quercetin), tannin, and β -tocopherol (Lau et al., 2020).

Based on their nutritional values, bananas, and their by-products have demonstrated exceptional potential in a number of food industries. This article intends to provide an overview of the nutritional composition and bioactive compounds found in bananas and their by-products. It also aims to discuss its application in the food industry.

Contribution: In this work, we have revealed several innovative aspects of bananas and their products including nutritional properties, which can be utilized as a sustainable source of income for the agriculture industry. Moreover, the contribution of this work can be summarized as:

- This work brings together several aspects of Banana and its by-products including its nutritional composition, bioactive compounds, and pharmacological benefits.
- Also, this work explores various daily life use cases and remedies of bananas and their by-products from the purview of the food industry.
- Moreover, this work demonstrates broader significance of the mentioned aspects, addressing various issues, that is, addressing waste, enhancing nutritional value, and contributing to sustainable food practices.

2 | NUTRITIONAL COMPOSITION OF BANANA AND ITS BY-PRODUCTS

The nutritional composition including proximate compositions, dietary fiber, vitamins, and minerals of the banana and its by-products is illustrated in Table 1. The composition varies depending on growth conditions, cultivars, and preparation process (Kraithong & Issara, 2021). Therefore, this section focuses on the different nutrients present in bananas and their by-products.

2.1 | Banana

The nutritional composition of bananas exhibits variation across different stages of maturity, particularly in

terms of sugar and carbohydrate content. At the green stage, bananas exhibit a notable abundance of starch and a comparatively diminished presence of sugar. The sugar content significantly increases while the starch content decreases during the full-ripening process (Evans et al., 2020). Starch is converted into sucrose via sucrose phosphate synthase, but starch is converted into non-reducing sugars by acid hydrolysis (Netshiheni et al., 2019). Carbohydrate variations play a significant role in the development of essential organoleptic characteristics in ripe bananas, including its sweet flavor and smooth surface (Evans et al., 2020). Consequently, moisture content increases with increased storage temperature due to osmotic flow of water and respiration breakdown of starch. Being a rich source of minerals, mainly potassium (K), it is also beneficial for human health (Anyasi et al., 2018). It is used for muscular contraction control. Different varieties of banana deliver macro and micro minerals like potassium, calcium, phosphorus, sodium, magnesium, zinc, iron, manganese, boron, and copper (Qamar & Shaikh, 2018). It is widely recognized that, apart from genotype, abiotic variables and agricultural methods play a significant role in shaping the mineral composition (Borges et al., 2020). Mature bananas contain relatively high contents of vitamins A (carotene), C (ascorbic acid), and B-complex (niacin, thiamine, and riboflavin) whereas they are moderately rich in vitamin B6, that is, pyridoxine (Qamar & Shaikh, 2018). During banana fruit ripening, the protein content is raised from 1.3 to 1.8 gm/100 gm FW and 4–7 gm/100 gm DW (Pareek, 2016). Among the proteins in unripe bananas, chitinase enzymes were the most abundant, with two isoforms present in the ripe fruit. At the climacteric stage, isoflavone reductase is abundant (Toledo et al., 2012). During banana fruit ripening, malate dehydrogenase, starch phosphorylase, and pectate lyase are formed (Netshiheni et al., 2019).

2.2 | Banana peel

The chemical nature of fruit peels renders them an often-utilized nutrient supplement. Similar to its pulp equivalent, the composition of a BP is characterized by a significant amount of organic matter, including fiber, lipids, proteins, and carbohydrates. Additionally, it serves as a prominent reservoir of diverse bioactive compounds that exhibit a range of functional properties (Zaini et al., 2022). Studies documented fresh BPs to have less moisture and fat content compared to fermented peels. The shelf life, freshness, and enzyme activity in fresh peel is basically responsible for metabolic processes, which causes rapid degradation, thus leading to reduction in its moisture content (Hassan et al., 2018; Romelle et al., 2016). Similarly, the reduced fat content of the fresh BPs contributes significantly to the energy content of feeds along with increasing in the storage

TABLE 1 Nutritional composition of banana and its by-products.

Components	Banana	Banana peel	Banana leaves	Banana pseudo-stem	Banana blossoms
Proximate composition (%)					
Moisture	77.19	8.28	66	5.0–8.9	90.01–90.23
Protein	0.3	4.77	14.98	2.2–5.0	1.99–1.43
Ash	0.1	15.30	10.37	1.8–7.8	3.21–2.42
Fat	NA	13.15	21	NA	0.43–0.54
Carbohydrates	22.12	9.4	60–71	20.5–38.8	95.23–95.61
Minerals (mg/100 g)					
Potassium	308–426	4.39	3.8	10.63	553.3
Phosphorus	15–29	211.3	1.8	2.09	NA
Calcium	4–7	59.1	1.2	4.01	56.0
Copper	0.025–0.186	0.51	0.00797	0.02	NA
Zinc	0.11–0.24	0.033	0.0322	16.60	NA
Iron	0.19–0.41	47	0.2778	30.65	56.17
Manganese	0.116–0.829	0.702	0.2566	27.86	NA
Vitamins (mg)					
Vit. A	NA	3.21	NA	NA	NA
Vit. B1	0.062	1.79	NA	0.15	NA
Vit. B2	0.072	NA	NA	0.08	NA
Vit. B3	0.0672	NA	NA	0.73	NA
Vit. B6	0.0242	2.93	NA	0.33	NA
Vit. C	18.4	NA	NA	NA	NA
Vit. E	NA	1.03	NA	NA	NA
References	Ranjha et al. (2022)	Hikal et al. (2022), Mosa and Khalil (2015)	Mat et al. (2020)	Mydhili et al. (2022), Ramu et al. (2017)	Arya Krishnan and Sinija (2016)

capacity, thus diminishing the rancidity development (Hassan et al., 2018; Okareh et al., 2015). The lower ash content obtained from fermented peels compared to that of unfermented peels is an indication of low mineral content of later one (Ozabor et al., 2020). The increased fiber content in fermented peels, as opposed to unfermented samples, leads to the elimination of possible mutagens, xenobiotics, and steroids by being bound to dietary fiber molecules. Consequently, this process enhances an individual's digestion (Eleazu et al., 2013; Romelle et al., 2016). BP flour has also been reported to contain high protein content compared to that of BPs. The reason for the same is attributed to varietal variance, analytical methodologies, agro-ecological situations, or time gaps between harvesting and evaluation (Khatun et al., 2021). They are also reported to be abundant in minerals such as (phosphorus, iron, calcium, magnesium, and sodium) and vitamins such as (Vitamin D, ascorbic

acid, and pyridoxine) (Hikal et al., 2022). Ripe BPs have also been reported to contain less starch content compared to unripe BP (Sogi, 2020). The reason for the same is attributed to reduction in enzyme-resistant starch and an increase in water-soluble pectin during the stage of ripening (Phillips et al., 2021).

2.3 | Banana leaves

In a study conducted by Mat et al. (2020), it was revealed that the moisture content of fermented banana leaves was significantly higher (68.55%) compared to that of fresh leaves (66.05%). The cause of this phenomenon is ascribed to an excessive presence of moisture during the process of fermentation. In a similar manner, Gregory et al. (2016) conducted an experiment wherein fresh banana leaves were sliced, dehydrated at a temperature

of 40°C, and afterward provided as feed to sheep that were experimentally infested with parasites. It was shown that the tannins present in banana leaves prevent serpentine woolly nematode eggs, providing parasite resistance. As a result of receiving a meal containing 50 g of banana leaf powder per kilogram, all characteristics of broilers in the banana leaf treatment group improved. This was compared to those in the control group. The reason for the same might be the fact that banana leaves comprise of numerous nutrients and huge levels of fructo-oligosaccharides, which further enhance beneficial bacteria growth in the intestinal tract of broilers and produce vitamins and digestive enzymes, thus improving broiler absorption (Zou et al., 2022). Magnesium content was found to be highest in banana leaves followed by potassium, sodium, phosphorus, and calcium, that is, 9500, 3800, 1200, 2000, and 1800 ppm/DW, respectively. Variations in nutrient content may be due to fertilization, soil type, or banana species (Mat et al., 2020).

2.4 | Pseudo-stem

Researchers studied the moisture content of different varieties of BPS that include Poovan, Nendran, Monthan, Karpooravalli, Ney Poovan, Grand Naine, and Red banana. Out of all, the Karpooravalli and Poovan varieties had the highest and lowest moisture contents, 8.9% and 5.0%, respectively. The variation in the moisture content might be caused by the binding ability of the finished products due to the presence of high molecular weight compounds like cellulose, lignin, and hemicellulose (Mydhili et al., 2022). The maximum ash content was observed in the Ney Poovan (7.8%) and the lowest was in the variety Poovan (1.8%). A high ash concentration suggests that the sample contains more minerals (Aziz et al., 2011). The variety Monthan (5.5%) had the highest protein content whereas the variety Poovan (2.2%) had the lowest. Usage of pseudo-stem protein can be done for production of low-gluten foods such as cake (Go et al., 2021). Carbohydrate content in banana pseudostem has a low glycemic index and thus can be utilized as a flour substitute in functional foods. The total carbohydrate content was highest in the Karpooravalli variety (38.8%) while the Red Banana variety had the lowest carbohydrate content (20.5%) (Mydhili et al., 2022). It can be attributed to the different parts of the pseudo-stem used or the drying method used (Ramu et al., 2017).

2.5 | Banana inflorescence

Banana inflorescences are commonly eaten in the form of curry, fried, or as a boiled salad along with rice and wheat bread. Besides being freshly consumed, it can also be incorporated into a variety of food products like

dehydrated vegetables, pickles, and canned food (Mostafa, 2021). The proximate composition of banana inflorescences has been widely evaluated (Basumatary & Nath, 2018; Ramu et al., 2017; Suffi et al., 2021). The inflorescence of bananas contains a significant amount of carbohydrates and proteins, with a low-fat content and a high moisture level, indicating a short shelf life. Banana inflorescences is considered a rich source of beneficial unsaturated fatty acids like α -linolenic, oleic, and linoleic acids (accounting for more than 60% of total fatty acids) (Lau et al., 2020). Similarly, other studies found that linoleic acid dominated among all the fatty acids present in the banana inflorescence. It accounted for 84.80% of the total fatty acid concentration. Thus, it is also considered a good source of unsaturated fatty acids that can reduce the risk of heart disease (Ramu et al., 2017). Banana inflorescence has a balanced mineral composition.

Potassium (K) is the macro-element with the highest concentration, followed by calcium (Ca) and magnesium (Mg). On the other hand, banana inflorescence has a high level of iron (Fe) among its microelements (Basumatary & Nath, 2018; Ramu et al., 2017). Genetic differences between cultivars and environmental growth conditions may also contribute to banana inflorescence nutritional variability (Lau et al., 2020).

3 | BIOACTIVE COMPONENTS IN BANANA AND ITS BY-PRODUCTS

Banana is considered as a rich source of several bioactive compounds such as phenolics, carotenoids, glycosides, alkaloids, phlobatannins, tannins, terpenoids, saponins, steroids, biogenic amines, and phytosterols (Pereira & Maraschin, 2015; Singh et al., 2016). Due to their health-beneficial effects, these compounds are highly required in the diet (Netshiheni et al., 2019). Due to their antioxidative activities, these compounds are beneficial in protection of the body against oxidative stress (Padam et al., 2014), controlling gene expression in cell explosion and apoptosis and significant in controlling blood pressure as illustrated in Figure 2. Therefore, this section focuses on the bioactives present in bananas and their by-products.

Nutraceuticals and bioactive compounds derived from plant sources often require extensive research and development, as well as various stages of testing before being marketed safely (Figure 3). Compounds found in banana by-products have been commercialized in the pharmaceutical industry as nutraceuticals (Zhang et al., 2005).

According to recent research, banana by-products such as green culled bananas, peels, and pseudo-stems may be a valuable raw material and a cheap source of high-quality pectin, starch, and cellulose for the food industry. The various chemical compounds extracted from banana by-products are listed in Table 2 (Zhang et al., 2005).

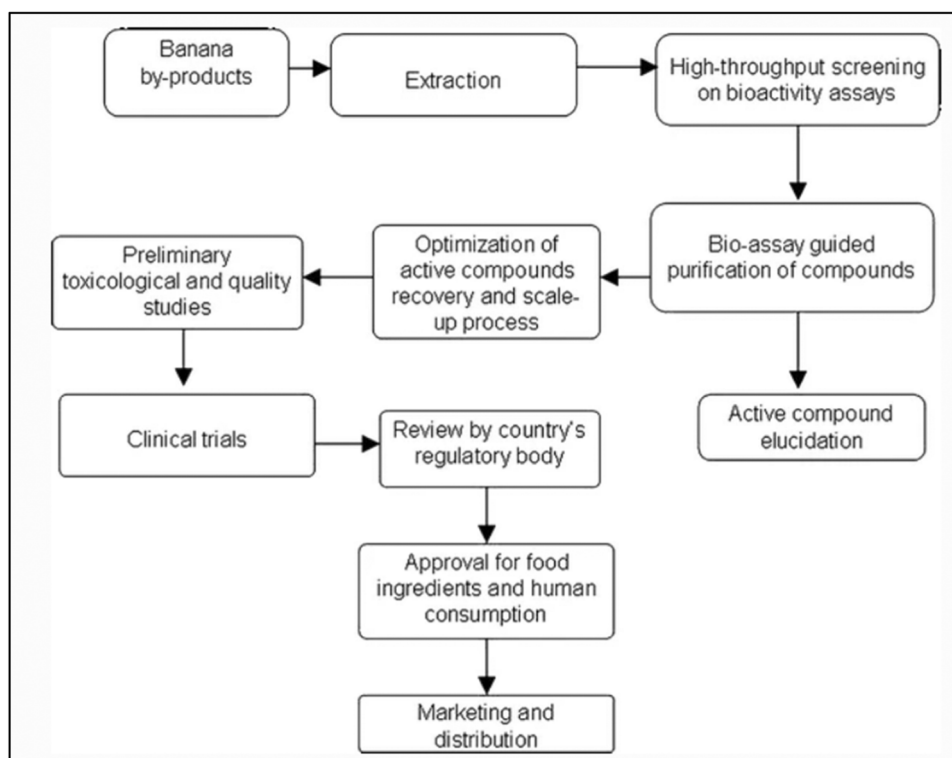


FIGURE 2 Extraction/determination of nutraceutical/bioactive components from by-products of banana (Zhang et al., 2005).

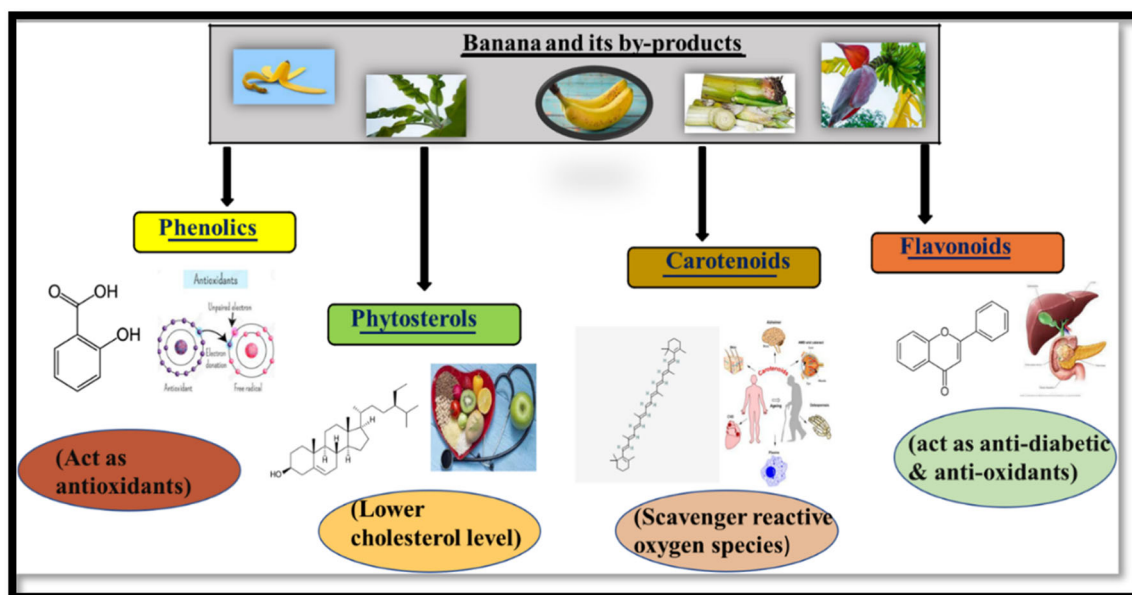


FIGURE 3 Bioactive compounds in banana and its by-products.

3.1 | Banana

In a comprehensive review, Singh et al. (2016) evaluated bananas as a rich source of bioactive compounds such as carotenoids, flavonoids, phenolics, amines, ascorbic acid, and vit. E that possesses antioxidant activities beneficial

for human health. Phenols are one of the major bioactive compounds in bananas that offer antioxidant effects and health benefits. On cutting bananas, polyphenols cause oxidative browning (Netshiheni et al., 2019). The total phenolic acid content in bananas varies from 11.8 to 90.4 mg/100 g, which imparts a bitter taste to unripe

TABLE 2 Starches, pectins, and cellulose from banana by-products for food application.

Chemical compound	Banana by product	Potential food application	References
Starches	Green culled banana (variety not specified)	Food thickeners, gelling agent, reinforcing agent in tablets, fillers	Aziz et al. (2011)
	Banana pith and pseudostem	Food thickeners, gelling agent, reinforcing agent in tablets, fillers	Pelissari et al. (2012)
	Unripe fruit	Viscosity and swelling properties	Sulaiman et al. (2011), Wang et al. (2012)
	Green banana	Substitution of cassava starch for textural properties of snacks (crackers)	Zamudio-Flores et al. (2006)
	Green banana plantain	Production of starch based edible films	Emaga et al. (2008)
Pectin	Banana peels (<i>Musa acuminata</i> AAA, <i>Musa acuminata</i> × <i>balbisiana</i> AAB)	Food thickeners, gelling agent	Adinugraha et al. (2005)
Cellulose carboxymethyl cellulose	Banana pseudostem	Food thickeners, gelling agent	Elanthikkal et al. (2010)
Microcrystalline cellulose	Not specified	Water retainer, reinforcing agent in tablets	Bhat et al. (2019)

bananas (Zamudio-Flores et al., 2006). Similarly, the most prominent classes of flavonoids reported in bananas are flavanols, such as kaempferol quercetin, myricetin, and cyanidins. They act as defensive scavengers against oxygen-derived free radicals and reactive oxygen species (ROS) liable for aging and diverse diseases (Singh et al., 2016).

Consumption of carotenoid-rich foods enhances immunity and diminishes the risk of diabetes, cancer, and cardiovascular diseases. Bananas are also a rich source of carotenoids. Carotenoids have many benefits, including their role as pro-vitamins and antioxidant properties, particularly the ability to scavenge singlet oxygen (Bhat et al., 2019). Karpooravalli variety of banana was found to contain the highest content of β -carotene (143.12 lg/100 g) with the highest aggregation of carotenoids in the nonedible portion (68 lg g/DW) (Singh et al., 2016).

Another bioactive compound, that is, phytosterols are classified as naturally occurring plant sterols that are employed as ingredients in functional foods because of their numerous health benefits. These benefits include minimizing blood cholesterol levels and lowering its absorption in the intestine (Singh et al., 2016). In banana pulp, free fatty acids and sterols are major phytosterols. In fact, bananas are an excellent source of phytosterols and steryl glucosides (Lopes et al., 2020). Villaverde et al (Villaverde et al., 2013). determined phytosterols in banana cultivars including cloeucalenone, cycloartenol, cycloeucalenol, campesterol, β -sitosterol, and stigmasterol. Vilela et al. (2014) analyzed chemical constituents of ripe banana pulp using GC–MS and identified phytosterol varies from 11.1% to 28.0% of the total amount of lipophilic compounds. Bhat et al. (2019), in

their study, determined that the difference in phytosterol content may be due to different varieties and environmental conditions.

3.2 | Banana peel

The BP scores the second highest level of phenolic content among all fruit peels, including pineapple, avocado, melon, papaya, and passion fruit. BPs contain the bioactive compounds such as flavonoids, tannins, phlobatannins, alkaloids, glycosides, anthocyanins, and terpenoids that have a number of pharmacological and biological effects such as antidiabetic, antibacterial, anti-inflammatory, and antihypertensive (Vu et al., 2018).

BPs contain many secondary metabolites, including tannins, gallic acid, catechin, anthocyanin, and epicatechin (Sidhu & Zafar, 2018). In addition, it has been observed that the peel of bananas possesses a significantly higher concentration of gallic acid, almost five times greater than that found in the pulp. This finding implies that BPs might be considered a rich source of antioxidants (Mohd Zaini et al., 2022). More than 40 compounds from BP have been identified as individual phenolics with a total amount of 47 mg of gallic acid equivalent (GAE)/g dry matter (DM). The phenolic components found in BP can be further classified into four subgroups, flavonols, hydroxycinnamic acids, flavan-3-ols, and catecholamines (Vu et al., 2018). In contrast to carotenoids and chlorophylls, BP antioxidant effects increased as it ripened and decreased when it became overripe. Hence, BP antioxidant activity is correlated with phenolic compounds rather than carotenoids and chlorophylls. For this reason, the maturation

stage must be considered depending upon its potential use (Vu et al., 2019). The high level of polyphenol in unripe BP makes them more effective than ripe BP extract at inhibiting human red blood cells oxidative hemolysis. Due to their physiological properties, carotenoids are beneficial to human health as they help in scavenging free radicals as well as reducing the progression of diseases, especially certain tumors and eye diseases (Bhat et al., 2019). Carotenoids recognized were β -carotene, lutein, β -cryptoxanthin, α -carotene, violaxanthin, auroxanthin, neoxanthin, isolutein and α -cryptoxanthin. The carotenoid content varied from 300 to 400 $\mu\text{g}/100\text{ g}$ as lutein equivalents in BPs by alumina column chromatography and HPLC (Singh et al., 2016).

3.3 | Banana leaves

Banana leaves exhibited a considerable content of total phenol, flavonoids, and tannin, that is, 27.95, 30.76, and 17.30 mg/g DW , respectively on treatment with acetone extract. Major phenolic compounds were ferulic and vanillic acids. On treatment with acetone extracts, banana leaves showed antimicrobial and antioxidant properties. They also showed significant activity against breast and hepatic cancer cells (Salama et al., 2020). Likewise, in another study, phytochemical screening of *M. balbisiana* banana leaves revealed flavonoids when treated with crude ethanolic extract (Yingyuen et al., 2020). Similarly, a significant number of phenolic compounds were found in banana leaves namely polyphenol oxidase which is reported to be beneficial in the treatment of Parkinson's disease (Bisht et al., 2016).

3.4 | Banana Pseudo-stem

According to Deng et al. (2020), BPS has a significant number of metabolites, particularly flavonoids, phenolic acids, amino acids, and alkaloids. As reported by Chakraborty et al. (2021), polyphenols and flavonoids such as ferulic acid, catechin, gingerol, and cinnamic acid have antioxidant properties. Ghany et al. (2019) revealed that pseudo-stems also showed antimicrobial and anti-tumor properties.

3.5 | Banana inflorescence

The presence of high phenolic content in the banana inflorescence signifies that these biological substances contribute to antioxidant properties, diminish ROS production and also induce free radical scavenging (Cicerale et al., 2012; Rao et al., 2016). Banana flowers have the highest total phenolic content of 1235.94 $\mu\text{g GAE/g}$ (Thaweesang, 2019). Flavonoids, an essential

secondary metabolite found in banana inflorescence, exhibit a multitude of biological activity, including anti-inflammatory, antioxidant, antifungal, antithrombotic, antiallergenic, anticarcinogenic, and hepatoprotective properties (Sumathy et al., 2011). Compared to the whole banana inflorescence, the male flower in the distilled water extract had the highest concentration of flavonoids (Rodrigues et al., 2020). It was observed that methanolic extract had the maximum flavonoid content at 0.25 mg QE/g (Rao et al., 2016). Banana inflorescence is also considered an extensive source of tannin, which is responsible for various biological activities such as antioxidant, anticarcinogenic, anti-inflammatory, and radical scavenging properties (Redondo et al., 2014). In a study, it was determined that banana flowers contain significantly higher tannin content than BPS ($86.87 \pm 02.43\text{ mg}/100\text{ g}$, $7.86 \pm 0.21\text{ mg}/100\text{ g}$) respectively (Ramu et al., 2017).

4 | PHARMACOLOGICAL IMPORTANCE OF BANANA AND ITS BY-PRODUCTS

Bananas and their by-products contain a lot of phytochemicals such as phenols, terpenoids, and saponins. Due to its pharmacological characteristics, researchers studied for its nutritional and therapeutic areas (Ajithkumar et al., 2017) (Table 3).

Variety of pharmacological activities such as antioxidant, antimicrobial, antidiabetic and so on are present in banana and its by-products (Ahmad & Danish, 2018) as illustrated in Figure 4. This section discusses the notable ones that are more significant in healthcare.

4.1 | Banana

4.1.1 | Antioxidant activity

The ripe banana has high antioxidant activity by 5% compared to the unripe banana (Aquino et al., 2016). Agronomical parameters mostly affect antioxidant activity in banana species. It has been reported that antioxidant levels in banana varieties such as Mas, Berangan, and Raja vary between 3.2% and 63.1% (Shian & Abdullah, 2012). As a result of differences in phenolic compounds, bananas contain a higher ABTS (2,2'-Azino-bis-3-ethylbenzthiazoline-6-sulfonic acid) than nonclimacteric fruits and a lower ABTS than apples, lemons, plums, and strawberries (Darsini et al., 2012). The antioxidant activity of banana pulp extract varied with solvent polarity and total phenolic content. The evaluation of banana pulp with ethyl acetate showed higher scavenging activity than the ethanol extract (Jain, 2011). Also, the types of solvent used for extraction affect banana antioxidant activity.

TABLE 3 Pharmacological activities of banana and its by-product.

Banana and its by-products	Antioxidant activity (µg/mL)		Antimicrobial activity (mm)		Anticancer activity (%)		References
	DPPH, ABTS		<i>Escherichia coli</i> , <i>Staphylococcus aureus</i> , <i>Bacillus subtilis</i>		HeLa, MCF-7, HepG2		
Banana	5.78–65.96, NA		14–25, 20–37, NA		NA, 12.07–61.21, NA		Ahmadi et al. (2019), Dahham et al. (2015), Jahan et al. (2019), Mondal et al. (2021), Shian and Abdullah (2012)
Banana peels	90.28–113.09, 12.89–75.39		09.67–18.15, 10.53–19.57, 10.66–20.60		44.1–62.3, 8.86–75.3, 22.18–56.09		Aboul-Enein et al. (2016), Barroso et al. (2022), Durgadevi et al. (2019), Hasanin et al. (2021), Mondal et al. (2021)
Banana leaves	27.81–40.63, 45.76–134.62		09.87–18.15, 09.00–22.00, 09.00–21.32		16.75–54.35, 48.60, 33.50		Deep et al. (2021), Salama et al. (2020)
Banana pseudo-stem	76–122, NA		8.29–22.14, 7.16–14.84, 9.63–14.33		82–284, NA, 1.28–79.06		Abdel Ghany et al. (2019), Kumar et al. (2014), Mondal et al. (2021), Ramu et al. (2015), Saravanan and Aradhya (2011)
Banana inflorescence	2.2–25.10, NA		8.92–18.6, 9.39–15.24, 7.19–14.53		16.2–83.8, NA, NA		Das et al. (2020), Marikkar et al. (2016), Mondal et al. (2021), Ramu et al. (2015)

4.1.2 | Antimicrobial activity

The aqueous and ethanolic extracts of unripe bananas exhibit high antibacterial activity against both gram-positive and gram-negative bacteria. The unripe banana has minimum bacterial concentration (MBC) and minimum inhibitory concentration (MIC) ranging from 32 to 512 mg/mL in ethanol and 20,512 mg/mL in aqueous solvent (Imam & Akter, 2011). In another study, Kurdi gum (KG) and Farsi gum (FG) were applied as coatings to banana fruits to evaluate their antimicrobial activity. Only the farsi gum coating did not exhibit any microbial activity against *Escherichia coli*, *Bacillus cereus*, *Salmonella typhimurium*, and *Staphylococcus aureus*. The reason for the same is attributed to essential oils present in Kurdi gum which lead to antimicrobial activity against several bacterial strains due to the presence of α - and β -pinene (Sharifi & Hazell, 2011). It has been established that phenolic compounds are responsible for the antimicrobial properties of herbs (Rezaei et al., 2018). The aqueous extract of banana did not show antimicrobial activity, whereas methanolic and ethanolic extracts are effective against pathogen species (Jahan et al., 2019). Similarly, the antibacterial effect of banana extract is less effective against gram-positive bacteria than gram-negative bacteria (Karadi et al., 2011).

4.1.3 | Anticancer activity

The banana extract depicted anticancer activity against human colorectal carcinoma cells (HCT-116), Human Umbilical Vein Endothelial Cell (HUVEC), and human hormone-sensitive and invasive breast cancer cell line (MCF-7). As nonpolar solvents have high anticancer activity, the polarity of the solvent used for extraction is considered to have an effect on the anticancer property. Hexane extract showed the highest cytotoxicity against MCF-7 and HCT-116 with 61.21 and 62.04 inhibition (Dahham et al., 2015). Banana Hexane extract exhibit the highest anticancer activity in all cancer cell lines. However, the most significant inhibition was observed in PANC-1 cells with a value of 510.036 µg/mL against MCF-7/SC (stem cell-like) breast cancer cells, human hepatoblastoma (HepG2) cells, human breast epithelial cells (MCF10A) (Kim et al., 2022). Yellow banana fruit exhibited effective anticancer activity against human cancer cells (Ht-29) due to its tumor necrosis factor (TNF) which destroys cancer cells (Praveena et al., 2018).

4.2 | Banana peel

4.2.1 | Antioxidant activity

It was reported that the methanol (45.76%) and acetone (46.63%) extract of BP showed higher antioxidant activity

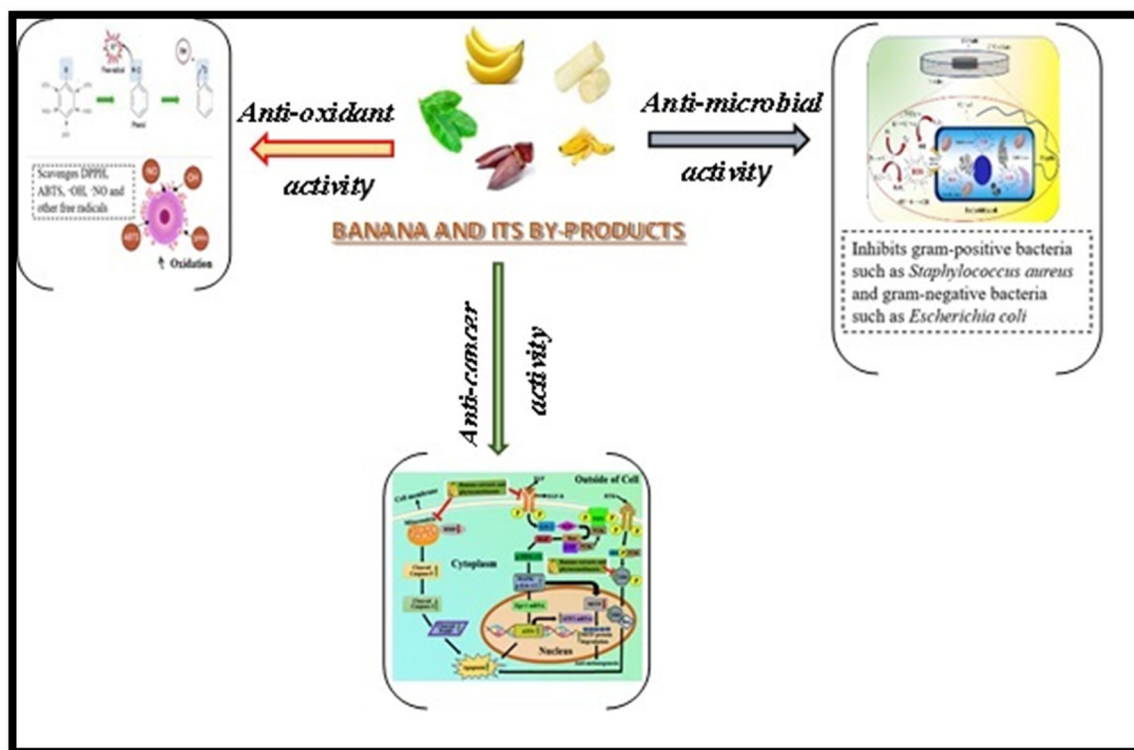


FIGURE 4 Pharmacological properties of banana and its by-product.

than the aqueous (37.85%) and ethanol extract (40.45%) due to the presence of high levels of flavonoids and total phenolic content (Aboul-Enein et al., 2016). In similar ways, BP flour can be tested for antioxidant activity using ABTS (2,2'-Azino-bis-3-ethylbenzthiazoline-6-sulfonic acid), ferric-reducing antioxidant power (FRAP) and oxygen radical absorbance capacity (ORAC). It is observed that hydrogen atom transfer (HAT) antioxidant activity is better than ABTS and FRAP (Rebello et al., 2014). The antioxidant activity of BP is affected by the ripening stages, as ripened BPs contain the highest antioxidant activity than green ones (Pertuzatti et al., 2014). On the other hand, it was observed that the high concentration of dopamine, catecholamines, and L-dopa in banana extracts provides antioxidant activity (González-Montelongo et al., 2010). Due to the higher FRAP activity, BPs have a better ferric reduction capacity than pulps (Durgadevi et al., 2019).

4.2.2 | Antimicrobial activity

Ethanol extraction by sonication method produced better results due to superior extraction of bioactive components from banana (Aboul-Enein et al., 2016) and kinnow peel powder (Safdar et al., 2017). Similarly, BP oil extracted from methanol had high antibacterial activity against bacterial species due to the antimicrobial component 2-methyl-5-(1-methylethyl) phenol (Mordi et al., 2016). It was exhibited that 50% ethanol extract

of BP powder by sonication method achieved the highest bacterial inhibition at 600 ppm against different gram-positive bacteria, particularly *E. coli* (15.43 mm), *P. aeruginosa* (12.11 mm), *S. cerevisiae* (13.21 mm), and *S. aureus* (11.31 mm). The most favorable results obtained by ethanolic extract and sonication assay may be due to the better extraction of bio-active components than other solvents and techniques (Chaudhry et al., 2022). Acetone and ethanol extract of peels were examined by a well diffusion method against different microorganisms and acetone extract exhibited maximum inhibition. A number of factors influence the antimicrobial property, including the bioactive compounds, plant parts, and the polarity of solvents (Aboul-Enein et al., 2016).

4.2.3 | Anticancer activity

In the aqueous methanolic extract of BP, bioactive components were found to have a significant amount of cytotoxic effect against MCF-7 breast cell lines, which prevented the growth of cancer cells (Durgadevi et al., 2019). The BP contains ferulic acid, which acts as an anti-melanogenesis agent by stimulating growth factors in the vascular endothelium and regulating nitric oxide production (Phacharapiyangkul et al., 2019). BP flavonoids inhibit ROS, hindering the cell cycle, and destroying tumor cell production (Mohd Zaini et al., 2022).

In another study, the hydroalcoholic extract of BP was examined for anticancer activity against hepatocellular carcinoma (HepG2), breast carcinoma (MCF-7), colorectal adenocarcinoma (Caco-2), and malignant melanoma (A-375). Of all cell lines, (A-375) was the most sensitive, whereas Caco-2 was the most resistant to hydroalcoholic extract (Barroso et al., 2019). The plant-derived polyphenols are more susceptible to inhibiting the growth of tumors and make cancer cells more sensitive to chemotherapy and radiation treatment (de Souza Vandenberghe et al., 2017).

4.3 | Banana leaves

4.3.1 | Antioxidant activity

The scavenging ability of the methanol extract of banana leaves was higher (44.50 µg/mL) than that of the acetone extract (27.81 µg/mL). It was suggested that the content and presence of phenolic compounds were directly related to the antioxidant activity of banana leaves (Gaafar et al., 2015). Banana variety and solvent type affects the Antioxidant activity of banana leaves. *Musa balbisiana* leaf extract exhibits higher Antioxidant activity than *Musa acuminata* (Handayani et al., 2021). Banana leaves have physical and chemical properties that are affected by the drying temperature. The high-temperature dried leaves show a low antioxidant activity compared to fresh leaves (Sagrin & Chong, 2013). According to researchers, the trolox equivalent antioxidant capacity (TEAC) value of banana leaf aqueous extract was 24–295 times lower than that of acetone and methanol extracts, respectively. Similar to avocado leaves, different extraction procedures and extraction solvents could account for significant variation in antioxidant activity. Different extraction procedures and extraction solvents could account for significant antioxidant activity variation. The TEAC value of banana leaf aqueous extract was 24–295 times lower than that of acetone and methanol extracts (Gopalan & Tyug, 2021).

4.3.2 | Antimicrobial activity

Banana leaf extract had maximum inhibitory value against *P. oxalicum* (40%) compared to minimum value against *A. alternate* (10%). The MIC value of banana leaf extract was recorded against *B. subtilis* (10 mg/mL), *P. aeruginosa* (30 mg/mL), *E. coli* (7.5 mg/mL) and *S. aureus* (15 mg/mL), respectively (Bisht et al., 2016). The solubility degree of bioactive compounds in solvents affects antibacterial activity. Ethanol extract of banana leaf showed better antibacterial properties against *S. aureus* and *E. coli* while chloroform extract proved better against *B. subtilis*, *P. aeruginosa*, and *S. aureus* (Naikwade et al., 2014). Due to the presence of phenols, tannins, saponins and flavonoids, ether extract of leaf showed highest antimicrobial activity

toward gram-positive bacteria (*B. subtilis*) than gram-negative (*S. aureus*) (Yadav et al., 2021). The concentrations of 40 and 60 mg/mL banana leaf extract were most effective against *T. mentagrophytes*, which exhibited a 5 mm diameter inhibition zone. Nystatin was utilized as a positive control against *T. mentagrophytes*, which showed lesser effects on the colonies of fungi by producing a 2 mm zone (Harith et al., 2018).

4.3.3 | Anticancer activity

Banana leaf extract contains critical micronutrients and phytochemicals, which prevent cancer by inhibiting the cell growth and antioxidant activity (Dahham et al., 2015; Salama et al., 2020). Similarly, banana leaf silver nanoparticles (BL-AgNPs), at low concentrations (100 g/mL), were effective in cancer cell suppression, with a range of 54.187% and 65.513% respectively. The newly developed anticancer drug (BL-AgNPs) destroys both cancerous and normal cells. The primary factor contributing to the superior cytotoxic impact (%) of BL-AgNPs on the MCF 7 cell line when compared to banana leaves extract is the significant interaction between the bio-green compounds present in the banana leaves extract and the silver ions in the sample (Raghavendra et al., 2021). Additionally, eugenol and phytol exhibit the highest antioxidant and anticancer properties. Banana leaf extract with 320 g/ml concentration inhibits A375 cancer cells (Bezerra et al., 2017; Deep et al., 2021).

4.4 | Banana Pseudo-stem

4.4.1 | Antioxidant activity

The scavenging activities of BPS vary with the type of solvent used for inhibition of free radicals. The acetone (70.47%–23.64%) has high scavenging property compared to methanol (40.02%–12.73%) and ethyl acetate (28.55–5.35) (Saravanan & Aradhya, 2011). Additionally, increased extract concentration results in increased DPPH activity. The lower IC₅₀ (2.2 mL) indicates stronger antioxidant activity (Abdel Ghany et al., 2019). The polyphenol activity of the “Nendran” variety had higher DPPH radical scavenging activity (80.86%) at a concentration of 100 g/mL than the “robusta” (69.84%) and Kathali (49.01%) (Anusuya et al., 2013). Consequently, banana pseudo stem extract exhibits antioxidant properties due to its high phenolic component concentration (Aziz et al., 2011).

4.4.2 | Antimicrobial activity

The aqueous extract of banana pseudo stem possesses strong in-vitro antimicrobial properties. The aqueous extract of banana pseudo stem exhibits high inhibition

against streptococcus sp. (21 mm) compared to methanolic extract (14 mm) (Onyema et al., 2016). As a result of the presence of saponins, tannins, alkaloids, flavonoids, phenols, and glycosides in banana pseudo stem extracts, they possess antimicrobial properties against gram-positive and gram-negative bacteria (Ramu et al., 2015). The hexane extract of banana pseudo stem showed the maximum inhibitory effects against gram-negative bacteria (*E. coli*, *P. aeruginosa*, *S. typhi*, and *E. aerogenes*) as compared to the chloroform, ethyl acetate, and acetone while the chloroform extract has the highest inhibition against gram-positive bacteria as compared to the hexane, ethyl acetate, and acetone (Kandasamy et al., 2020).

4.4.3 | Anticancer activity

The banana pseudo stem extract containing flavanol and flavones is cytotoxic against the cancer cells MCF-7 (breast cancer) (Gupta et al., 2022). Based on the presence of glycosides, alkaloids, phenols, flavonoids, and saponins in banana pseudo stem extract, it showed higher cytotoxicity against HCT-116 (86.53) and HepG-2 cell lines (79.06) compared with other banana pseudo stem extracts (Abdel Ghany et al., 2019; Ogbonna et al., 2016). An analysis of the cytotoxicity activity of the drug against PA1 cells was conducted and compared with the standard (Cisplatin drug). To test the anticancer properties of synthesized silver nanoparticles, several concentrations of silver nanoparticles have been used, including 50, 100, 150, 200, 250, and 300 g. According to the results, the anticancer activity of the molecules increased with the concentration of AgNPs (antitumor activity of biogenic) and the viability of the cells decreased (Kumari et al., 2022). However, the exact mechanisms underlying the anticancer activity of banana pseudo stem extract are not yet fully understood and require further investigation. The presence of various bioactive compounds in the extract is believed to contribute to its cytotoxic properties. These compounds may interfere with cancer cell growth, induce apoptosis, or inhibit angiogenesis.

4.5 | Banana inflorescence

4.5.1 | Antioxidant activity

The banana flower extract from ethanol (88.62%) with a 63 µg/mL IC₅₀ value exhibited high antioxidant activity with maximum inhibition (Joseph et al., 2014). The flowery ethanol extract of the *Poovan* variety showed higher antioxidant activity (82%) than the *Monthan* variety (Arya Krishnan & Sinija, 2016). Antioxidant properties depend on the total phenolic content of flower extracts. The *pisang susu* variety had maximum ABTS radical scavenging effects (24.73 ± 0.03 µmole of Trolox equivalent/g of extract) whereas, *pisang rastali* variety had minimum ABTS value (3.72 ± 0.09 µmole of Trolox

equivalent/g of extract) (Marikkar et al., 2016). Extraction techniques and agronomical parameters affect banana flower extract antioxidant levels (Hue et al., 2012).

4.5.2 | Antimicrobial activity

Antibacterial activity of banana flower extract toward gram-negative and gram-positive bacteria was examined by the development of inhibition zones around the wells. Higher inhibition zones were detected on the plates with 20 µg/mL of Ampicillin added to the methanol extract. *S. aureus* exhibits a larger diameter than banana flower extract (10 mm), whereas other bacterial strains show a diameter range of 5–8 mm. In the agar-well diffusion method, the inhibition zone is formed due to bioactive components in the flower extract (Das et al., 2020). All banana flower extracts inhibit gram-negative bacteria, such as *S. flexneri*, bacteria *K. pneumonia* and *E. coli* except ethyl acetate and aqueous extract shown mild impact to inhibit *K. pneumonia* (Govindaraj et al., 2022; Ramu et al., 2015). The alkaline extract of banana flowers has antibacterial activity at a lower dose (5 and 10 mg/mL). *S. aureus* (gram-positive) was found to be more susceptible to protein extracts than *E. coli* (gram-negative) (Siththiya et al., 2018).

4.5.3 | Anticancer activity

The ethanolic extract of banana flower showed anticancer activity against HeLa cervical cells. The ethanolic extract inhibited cell proliferation and induced apoptosis (Nadumane & Timsina, 2014). It has been demonstrated that banana flower extract contains anticancer properties against cancer cells, which led to the reduction of human colorectal adenocarcinoma cell line (HT29) colon cancer cells after treatment (Govindaraj et al., 2022). A banana flower alcoholic extract shows a much greater efficacy against breast cancer, HeLa cells, HT29 cell line, and colorectal cancer cells (HCT116) (Chowdary et al., 2022). The banana flower extract contains phenolics and flavonoids that cause cell death and apoptosis (Lau et al., 2020). The methanol extract of banana flower exhibited anticancer and antiproliferative potential against HeLa cells (Das et al., 2020). Syringic acid, gallic acid, ferulic acid, and catechol are present in banana methanolic extracts, which exhibit apoptotic properties (Arun et al., 2017).

5 | APPLICATIONS OF BANANA AND ITS BY-PRODUCTS IN FOOD INDUSTRY

In addition to banana flour, banana chips, sauce, patties, cookies, yogurt, biscuits, chicken products, nut chocolate, banana plant parts also serve as packaging materials

and as foods as shown in Figure 5. The physiochemical and organoleptic properties of the finished product depend on the plant cultivar, the percentage of the plant used and applied to food products.

5.1 | Banana

5.1.1 | Banana flour

The utilization of banana flour as a functional ingredient has received attention due to its high content of dietary fiber, resistant starch, and potassium (Pragati et al., 2014). Banana is also considered beneficial in the treatment of various health ailments, such as maintaining a healthy digestive tract, protecting the kidneys, increasing muscle strength, preventing muscle cramps and depression (Christi et al., 2019). As a result, it has been recommended as an ingredient for the preparation of various bakery products such as cookies, breads, and pasta. The incorporation of banana flour in cookies by substituting 10% wheat flour resulted in it being highly acceptable with all quality features. It significantly enhanced the crude fiber (197.56%) and ash content (93.25%) (Alam et al., 2020). Similarly, the addition of whole green banana flour (WGBF) in bread formation enhanced macro minerals such as calcium, phosphorus sodium and magnesium with no substantial change in micro minerals namely iron, zinc, and manganese (Khoozani et al., 2020).

Gluten-free pasta made from raw banana flour was fused with protein and vegetable gums. After processing, it was observed that banana flour pasta was less prone to α -amylase digestion than pasta made from wheat flour (Zheng et al., 2016). According to Segundo et al. (2017), the inclusion of banana flour considerably increases the nutritional values of the cakes, by enhancing their polyphenols, antioxidant activity, and dietary fiber content (up to a three-fold enhancement in antioxidant properties). Yangilar (2015) analyzed green banana flour's effects on the physical, sensory, and chemical characteristics of ice cream. The findings revealed that ice cream's calcium content decreased while sulfur content increased. Organoleptic evaluation showed that the ice cream sample comprising banana flour (2%) was highly acceptable by the panelists.

5.1.2 | Banana breads

Breads developed from substitution of banana flour at different levels (5, 10, and 15) showed a decrease in bread volume and an increase in moisture, ash, and weight with an increase in the percentage of banana flour. Banana flour can be used by the bakery industry to make breads with enhanced fiber content and without significantly affecting sensory characteristics (Rahman et al., 2021). Similarly, air-oven dried (OD) and freeze-dried (FD) flour from WGBF were incorporated with wheat flour at 30% and showed

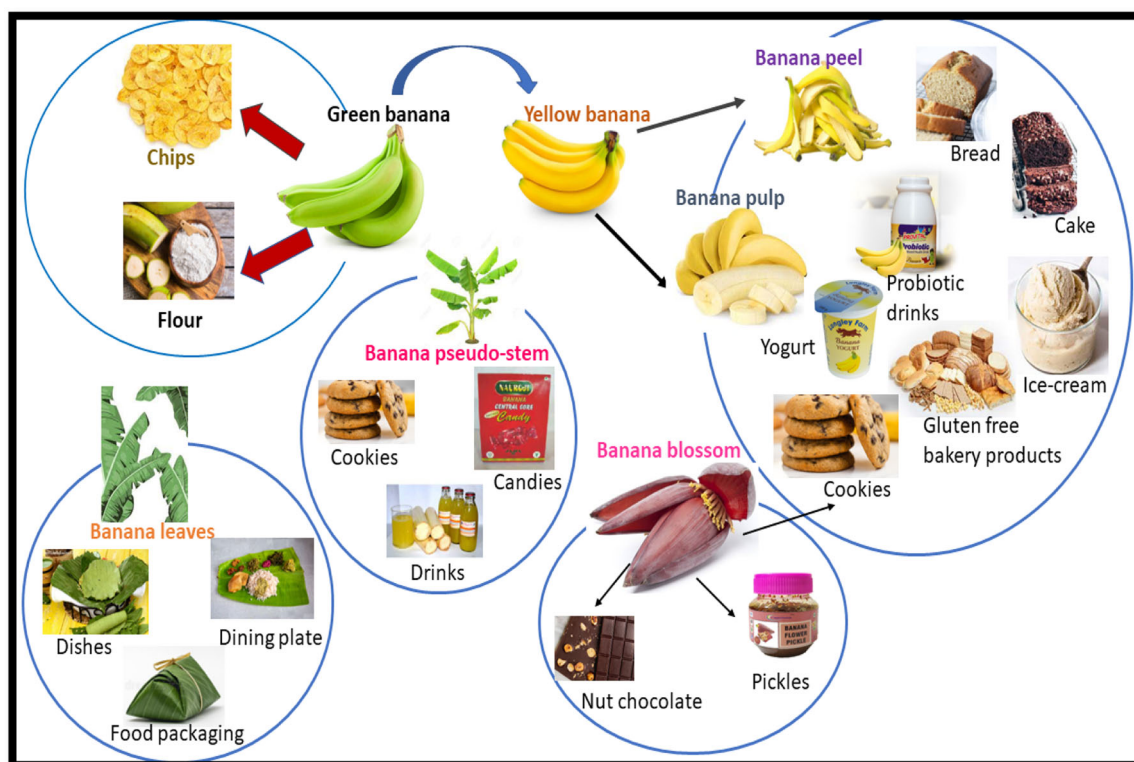


FIGURE 5 Applications of banana and its by-products in food industry.

higher elasticity, loss modulus, and complex viscosity of dough than 100% wheat flour bread ($p < 0.05$). Increase in WGBF incorporation resulted in harder, darker, and chewier bread (Amini Khoozani et al., 2020). Hernández-Aguirre et al. (2019) examined that incorporating water and pregelatinized banana flour into bread increased the size of gas bubbles (alveoli) and affected their diffusion. The size and diffusion of alveoli affect the texture and physical features of the bread. Hence, unripe banana flour can be utilized as a substitute ingredient to make gluten-free bread with superior quality features.

5.1.3 | Banana pasta

The green banana is considered to be a cheaper product with a lower industrial utilization rate. There was no notable difference between the fortified and control samples in terms of aroma, appearance, taste or overall characteristics after fortification with banana flour. There is potential for green banana flour to make gluten-free foods, which can increase the supply of products for people with celiac disease and provide a broader range of food choices (Zandonadi et al., 2012). The use of banana flour in pasta products showed low carbohydrate enzymatic hydrolysis, resulting in an increased variety of foods that have a low glycaemic index, available to the public. Furthermore, it produces high levels of indigestible components like nonstarch polysaccharides and resistant starch that benefit the gut (Ovando-Martinez et al., 2009). Additionally, an increase in banana powder concentration to produce pasta may increase the weight index (from 2.88 to 3.55) and cooking loss from 5.2% to 6.4%. It was revealed that banana powder moderately reduced the lightness of cooked pasta and had only a marginal effect on the color coordinates of uncooked pasta. With an increase in its concentration, the content of total phenolics and the antioxidant activity of pasta increased (Biernacka et al., 2020). Studies reported that pasta made with green banana flour was less susceptible to enzyme digestion than pasta prepared with wheat flour. This is because restricted enzymatic hydrolysis resulted in the formation of a gluten network in wheat flour pasta. Green banana flour pasta processing was affected by a protein network, but enzyme susceptibility was not altered significantly. This suggests that the lower enzyme vulnerability of pasta made from green banana flour may be due to the inherent enzyme-resistant properties of green banana starch (Zheng et al., 2016).

5.1.4 | Banana chips

Banana chips are popular worldwide and are placed in high-quality nutritional food products due to their low fat and calorie content. Its categorization is primarily based on ingredients, texture, and color (Ranjha et al., 2022). Similarly, banana chips were found to be

a healthy source of energy and minerals, namely phosphorus and potassium. Their preparation as a snack was determined to be easy and environmentally friendly (Abd Elmoneim et al., 2014). Wani et al. (2017) results showed that the longer frying time of banana chips resulted in a darker yellow color of the chips, particularly due to Maillard browning reactions. Additionally, a temperature of 160°C, time (2.69 min), and slice thickness (1.5 mm) were considered optimal conditions for the production of high-quality banana chips with an extended shelf life. Studies reported that the fatigue level of workers was diminished on supplementation of banana chips treated with the vacuum frying technique due to their high nutritional content (Christi et al., 2019). The incorporation of dried curry leaf (0.02%) and natural antioxidants that is, tocopherol acetate and sodium ascorbate (0.01%) in banana chips resulted in their better chemical and physical characteristics than untreated chips (Adrika et al., 2015).

5.1.5 | Banana puree/pulp

In addition to bakery products, bananas can also be processed to produce juice or dry powder by mashing them with or without water. Being a cheaper source of starch and sugar fillers, bananas can be added to other fruit juices to provide texture. Furthermore, the addition of this puree to mixed fruit smoothies acts as a potassium supplement (Mohapatra et al., 2011). Banana puree/pulp is susceptible to discoloration during processing and storage due to the activity of polyphenol oxidase (PPO) on phenolic compounds which can be diminished or inactivated by ascorbic acid and sodium metabisulphite (Walkling-Ribeiro et al., 2008). During puree preparation, it was found that puree made from ripening stage 6 bananas compared to that of ripening stage 5 bananas showed a higher starch content and viscosity value which resulted in its pleasant flavor and mouthfeel. This puree (ripeness stages 5 and 6) could be preserved for up to 14 days at 4°C, resulting in a long shelf life of banana puree which is profitable for the food industry (Yap et al., 2017). Similarly, in another research, green banana puree was utilized to substitute fat in pound cakes in different proportions (0%–100%), followed by reduction in sugar level (0%–50%). As per the check-all-that-apply (CATA), substitution of fat with green banana puree results in modifications of color, compaction, size of slice, and some characteristics of flavor, odor, and texture (Porto de Souza Vandenberghe et al., 2017).

5.1.6 | Fermented banana beverages

Bananas are technically used to produce fermented beverages such as wine and beer. Clarified banana juice,

low in acidity, rich in sugar and minerals is an appropriate substrate for the fermentation process (Mohapatra et al., 2011). Studies showed that post-fermentation of banana wine was better in taste, appearance, and aroma than primary fermentation and also contained higher brix degree, more total acid, alcohol, reduced sugar, and lower pH. In banana wines, ester and alcohol were the major volatile compounds (Chen et al., 2020). As fermentation proceeded, a gradual increase in alcohol content and titrable acidity of wine was observed (Nwobodo, 2013). Additionally, bananas can be used as a supplement in beer production for enhancing ethanol production volume in case of all malt-wort beer; which imparts beer a darker color especially due to browning enzymes. Crushed bananas and sorghum are mixed with water and allowed to ferment for about 3 days to produce a light beer with low alcohol content (2%–5%) (Shandilya & Siddiq, 2020).

5.2 | Banana peel

Extracts of BP may be used in food products as a natural preservative agent to increase the quality and shelf life of food due to their strong antioxidant and antibacterial characteristics. Different kinds of foods have been successfully fortified using BP extracts such as cookies and yogurt (Vu et al., 2018).

5.2.1 | Cookies

As a result of the study by Oguntoyinbo et al. (2021), the incorporation of BP flour in the cookies resulted in an increase in ash, crude fiber, and fat content. This varied from 2.26%, 1.03%, and 12.00%–2.73%, 1.33%, and 12.20%, respectively. Similarly, it also resulted in increasing their antioxidant properties, anti-nutritional value, mineral and total phenolic content without changing their physical properties. Alam et al. (2020) evaluated BP flour to have a higher water holding capacity (4.58 g/g dry sample) than wheat flour (3.19 g/g dry sample). Moreover, Arun et al. (2015), in their study revealed cookies to be a rich source of total dietary fiber after supplementation with BP flour. Shafi et al. (2022) reported that the increase in concentration of BP flour in cookies also increased their hardness and antioxidant activities. The DPPH, total phenolic content and lipid peroxidation were reported in the values of 47.761–70.298, 0.282–0.921 mg GAE/g and 36.74%–71.84% respectively and these values may reduce during storage period. Hence, findings revealed that cookies fortified with BP flour had high concentrations of phenolic content and antioxidant activities without affecting their nutritional and physical characteristics.

5.2.2 | Yogurt

BP extract was successfully used in bioactive compounds enriched yogurt. It was observed that an increase in BP extract concentration in yogurts resulted in an enhancement of its antioxidant activity and total phenolic content (Kabir et al., 2021). Added BP flour to yogurt significantly increased protein, ash, fat content, antioxidant activity, and total phenolic contents. Moreover, yogurt samples supplemented with BP flour demonstrated 12% higher sugar contents than nonfortified yogurt (Zahid et al., 2022). BP extract addition to yogurt considerably increases its shelf life by preventing nutrients loss (Chen et al., 2020).

5.2.3 | Cakes

Gluten-free and nutritious cake developed by addition of green BP flour with rice flour at different ranges such as 5, 10, 15, and 20%. Incorporation of 5% and 10% BP flour did not influence the specific volume, density, volume, and baking loss of gluten-free cake. It is expected that by developing a nutritious gluten-free cake alternative, individuals with celiac disease will have a wider variety of dietary choices (Turker et al., 2016). The substitution of wheat flour with BP flour to produce cake showed higher antioxidant properties. This was particularly evident in samples that incorporated 12% and 15% BP flour. As an outcome of this study, it is suggested that BP flour can be utilized up to 15% to complement and enhance cake quality characteristics, at a level that has numerous health benefits (Ahmed et al., 2021). Similarly, it was revealed that the incorporation of BP extract in cake development exhibits higher fiber and carbohydrates content than control cake (without BP powder) (Balakrishnan, 2021).

5.2.4 | Meat products

Banana peel powder (BPP) is considered a beneficial ingredient to incorporate into meat production due to its high nutritional value. Addition of BPP results in an increase in the dietary fiber, water holding capacity and cooking yield of sausages with a decrease in their fat content (Mohd Zaini et al., 2022). Inclusion of BPP to develop very low-fat chicken patties significantly increased moisture, ash content, and cooking yield along with the reduction in the emulsion stability, pH, water activity, fat, and cholesterol level compared to the control sample (Chappalwar et al., 2021). Additionally, chicken sausage incorporated with two different varieties of BP flours reported Saba BP flour to have poorer water, oil retention capacity and better swelling ability than Berangan BP flour. BP flour is recommended as a

value-adding material in functional chicken products due to its high nutritional status (Mohd Zaini et al., 2021).

5.3 | Banana pseudo-stem

5.3.1 | Biscuits/cookies

According to researchers, fortification of BPS in biscuits resulted in an increase in their protein, antioxidants, phytochemicals (flavonoids, phenols, alkaloids, tannins, and ascorbate) and ash content while decreasing their fat, moisture content, and viscosity compared to that of the control wheat flour biscuits (Chakraborty et al., 2021). Biscuits produced using BPSs provide a nutrient-rich product, particularly in terms of potassium, calcium, sodium, ash, fat, and fiber content (Maskey et al., 2020). Similarly, the nutrient profile of cookies fortified with BPS was found to be high in essential nutrients mainly protein, dietary fiber, fat, ash, iron, and calcium that is, 5.41, 7.28, 10.53, 0.21, 2.12, and 48.36 mg, respectively (Sree et al., 2022).

5.3.2 | Beverages/drinks

Nutraceutical beverages developed with incorporation of BPS resulted in high total flavonoid and phenolic juice content (12.59 mgCE/200 μ L and 58.11 mgGAE) (Saravanan & Aradhya, 2011). Similarly, the lemongrass-based isotonic drink was produced with BPS fortification. The pseudo-stem is considered a good source of calcium (Ca) which leads to 39.08 mg/100 g of the product. It was found that the drink has high carbohydrate content (11.70% per 500 mL of sample), whereas protein (0.06%) is the least abundant nutrient (Swarnalakshmi et al., 2019). Development of Ready to Serve products using BPS extraction is valuable because it is a rich source of calcium, sodium, potassium, and other minerals. According to the data, total sugar, moisture, protein, potassium, lipid, and calcium contents are 0.014%, 90%, and 0.005%, 874, 88, and 130 mg/l, respectively (Ranathunga & Jayasooriya, 2016).

Using dextranucrase and immobilized biocatalysts, BPS juice extract can be bio-transformed into a beneficial juice that can be used for other purposes. Bioprocessing of juice yielded oligosaccharides with significant resistance to enzymatic hydrolysis and demonstrated the ability to stimulate probiotic growth, representing its prebiotic properties and nondigestible nature. Furthermore, banana stem juice was recommended as a substitute for artificial culture medium for cultivation of *L. mesenteroides*, microbial bioprocessing, and dextranucrase enzyme production (Sharma et al., 2017). Another study exhibited that the inclusion of BPS in ready-to-drink beverages had noteworthy nutrient content such as carbohydrates, total sugar, sodium, zinc, copper, potassium, manganese and vitamin C ranging

from 13.92 to 16.08 g/100 mL, 10.62 to 10.83 g/100 mL, 0.920% to 0.935%, 1.034–1.117 ppm, 0.9183–0.9190 ppm, 8.13%–8.30%, 0.9544–1.0144 ppm and 0.673–0.707 mg/100 mL, respectively (Shiva et al., 2018).

5.4 | Banana inflorescence

Banana inflorescence extract can be used as a nutritional supplement for food products. As a result, their inflorescences or extracts can be used to enhance food quality. Though it has a substantial effect on the chemical, physiological, and organoleptic properties of food products, the proper amount of inflorescence or its extract to be added in different food products has been evaluated (Lau et al., 2020).

5.4.1 | Chicken products

According to the researchers, banana flower products formed at a 25% level displayed the highest preference in color, taste, aroma, texture, and overall acceptance. Additionally, it showed the highest antioxidant property (53.74% of DPPH radical scavenging activity) (Novidiyanto et al., 2020). However, its extract has also been used in burgers and sausages. Infusion of pork burgers was done utilizing approximately 2% hydroethanolic extract of banana inflorescence. Banana flower had no negative effects on the burger product's organoleptic and physiochemical characteristics (Schmidt et al., 2016). As banana inflorescence extracts prevent lipid oxidation during storage, they can also be used as natural antioxidants in meat products (Rodrigues et al., 2020).

5.4.2 | Biscuits/cookies/chocolate

In a study, pretreatment of banana blossoms was done with rice rinsed with water and it resulted in a higher content of fiber, protein, and minerals such as potassium, sodium, copper, magnesium, and iron in the biscuits compared to commercialized biscuits (Elaveniya & Jayamuthunagai, 2014). Incorporating banana flower to produce cookies showed a nutrient-rich profile, particularly fiber and iron, which helped produce red blood cells (Mishra et al., 2017). In a similar manner, banana flower can be utilized as a galactagogue to increase maternal milk production in lactating working women. According to this study, lactogenic biscuits (with banana flower flour added) significantly increased the volume of expressed breast milk compared to the placebo group (without banana flower flour) (Nordin et al., 2019). Incorporation of banana inflorescence in nutritious nut chocolate results in an excellent source of fiber, protein, potassium, iron, magnesium, and vitamin E (Komal & Kaur, 2019).

5.4.3 | Other uses

Similar to other parts of the banana tree, ethnobotanical studies revealed that banana flowers have been used for centuries to treat several diseases. In India, local food practitioners believe that the juice of banana flowers has beneficial properties for diminishing blood sugar level. “Ghrita” is a local dish that is prepared from banana inflorescence and roots, which is considered to be helpful in treating diabetes (Lau et al., 2020). The indigenous human from Brazil (Xokleng Indians), developed a syrup from banana inflorescence that is subsequently used as a medicine for the treatment of respiratory ailments (Correa et al., 2017).

5.5 | Banana leaves

The popularity of banana leaves as dining plates is attributed to their high content of polyphenols, which inevitably show antioxidant properties and improve digestion of the food by releasing its components namely potassium and vitamin C during serving of hot food. Due to the large blade size of the leaf, it accommodates different meals such as rice, curries, pachadi, chutney, sweet dish, and so on (Kora, 2019). Banana leaves were used as packaging material for dangke (local cheese from enrekang regency, south Sulawesi, Indonesia) and it exhibited less microbial contamination than plastic package (Zakariah et al., 2019). Furthermore, biodegradable packaging material was produced by combining banana leaves with bio-bleaching and bio-pulping techniques utilizing immobilized Xylanase enzyme beads. The sheets developed were coated with bioplastic, which enhanced their pliability (Jeenusha & Amritkumar, 2020). Banana leaves were employed as crucial packaging to prevent lulo fruit from losing quality and firmness during storage period. This study suggested that the banana leaves can be used to maintain the consistency of the fruit over an extended period of storage by reducing weight losses and color changes (Forero-Cabrera et al., 2017).

6 | CONCLUSION AND FUTURE PROSPECTS

Utilization of agricultural waste products toward the development of commercially viable food supplement for the population is considered a traditional concept. This, however, is treated as indispensable to maximize the use of available resources for reducing the exploitation of solid waste, neglecting the reduction of environmental pollution. Banana and its by-products, that is, peel, pseudo-stems, leaves, and inflorescences, not only vary in their composition but also unvaryingly comprise rich sources including dietary fiber, minerals, carbohydrates, and protein. In

addition to a wide range of potential uses, they are also regarded as beneficial raw materials for the food and nonfood industries. Owing to these advantages, the processing of banana and its by-products could lead to the production of healthy foods for the population, thanks to the presence of bioactive compounds in it that possess possible antioxidant, antimicrobial, and antidiabetic properties. Observing the aforementioned benefits, this work highlights the nutritional composition and bioactive compounds present in banana and its by-products. In fact, pharmacological studies already revealed its remarkable effect as an antioxidant, antimicrobial, and anticancer activity. Looking ahead, future perspectives open new research directions to identify specific bioactive compounds and their mechanisms of action. Innovative formulation and clinical studies are intended to optimize their incorporation into products, while industry adoption can drive sustainable practices. Assessing the environmental impact and promoting global outreach are expected to pave the way for a healthier and greener future.

ACKNOWLEDGMENTS

Not Applicable.

AUTHOR CONTRIBUTIONS

Payal Kumari: Writing—original draft. **Supriya S. Gaur:** Supervision; writing—review and editing. **Ravindra K. Tiwari:** Visualization; writing—review and editing.

CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

Data sharing is not applicable to this article as no data sets were generated or analyzed during the current study.

ETHICS STATEMENT

The authors have nothing to report.

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How to cite this article: Kumari, P., Gaur, S. S., & Tiwari, R. K. (2023). Banana and its by-products: A comprehensive review on its nutritional composition and pharmacological benefits. *eFood*, 4(5), e110. <https://doi.org/10.1002/efd2.110>